

Tsung-Yi Ma

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Senior AI Engineer with a strong mathematical background (NTU Math). Specialized in Generative AI, Mathematical Optimization (MIP/LP), and Real-time Voice Interaction Systems. Recognized as a Top 1% Performer (Global Role Model) at ASUS HQ for delivering high-impact AI projects.

EXPERIENCE

ASUSTEK COMPUTER INC.

Senior AI Engineer

AI Engineer

New Taipei City, Taiwan

Aug. 2025 – Present

Nov. 2023 – Aug. 2025

- **2025 Global Role Model Award:** Selected for the **Outstanding Contribution Award** (HQ selection, Top <1% rate company-wide) for exceptional performance and high-impact project delivery.
- **AI Interview Agent (GAI Recruiter):** Architected a real-time streaming speech recognition pipeline using **Python (Asyncio), Ray, and WebSockets**. Implemented a **sliding window approach** to achieve constant low latency (<500ms). Optimized inference parameters (VAD, beam size) to significantly reduce hallucinations.
- **Production Scheduling Optimization:** Designed mathematical models (MIP/LP) using **CPLEX** and **SCIP** solvers. Restructured constraints to reduce execution time from **24+ hours to <1 hour**. Simulations validated a **10% increase in production output** under raw material shortages compared to traditional FIFO methods.
- **License Plate Voice Recognition:** Fine-tuned **OpenAI Whisper** for specialized alphanumeric recognition, boosting accuracy to **>80%**. Implemented on-the-fly data augmentation (A-law encoding, noise injection) to ensure model robustness.

EDUCATION

National Taiwan University

Master of Mathematics

Taipei, Taiwan

Sep. 2021 – Jun. 2023

- **The Phi Tau Phi Scholastic Honor Society:** Inducted as an Honorary Member (Typically awarded to the top 3% of Master's graduates for academic distinction).
- **CS Foundations:** Data Structures & Algorithms (**A+**), Machine Learning (**A**). *Self-study:* Operating Systems, Computer Networks, Computer Architecture.
- **Thesis:** "Introduction to the Ising Model and the Ising Model with Disorders". Applied rigorous analysis to statistical physics models.

National Taiwan University

Bachelor of Mathematics

Taipei, Taiwan

Sep. 2017 – Jun. 2021

- **Research:** Selected for **NCTS Undergraduate Summer Research Program** (Noncommutative Probability).
- **Industry Seminar:** Selected for the **Hon Hai Research Institute** study group, exploring connections between Non-commutative Probability and Quantum Information.
- **Honors Math:** Completed Honors tracks in Analysis, Algebra, Geometry, and Complex Analysis (Advanced curriculum replacing standard core requirements).

Taipei Municipal Chien Kuo High School

Class of Science (Honors track for advanced scientific training)

Taipei, Taiwan

Sep. 2014 – Jun. 2017

PROJECTS

Distributed Whisper Streaming | Python, Ray, faster-whisper

[GitHub](#)

- Re-architected the system using **Ray** and **RayQueue**, decoupling stateful audio processing and **hypothesis alignment** from heavy GPU-bound inference.
- Solved the 1-to-1 memory bottleneck by multiplexing concurrent streams into a centralized GPU worker pool, maximizing resource utilization for production scaling.

TECHNICAL SKILLS

Languages: Python (Primary), C/C++, TypeScript/JavaScript, HTML/CSS, SQL

AI & Optimization: PyTorch, OpenAI Whisper, Ray, CPLEX, SCIP

Backend & Systems: FastAPI, Flask, Asyncio, WebSockets

Developer Tools: Git, VS Code, Linux, Tmux, Postman, Swagger, $\LaTeX$, uv

Code Quality: Ruff, Pyright, ESLint, Prettier